

Nishath Ahamed

Associate Software Engineer | Backend, GenAI and Agentic AI

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PROFESSIONAL SUMMARY

Associate Software Engineer with experience across backend engineering, enterprise fintech platforms, and GenAI application development. Strong foundation in Java/Spring, Python, FastAPI, SQL, Kafka, microservices, and modern agent engineering tools including LangChain, LangGraph, and LangSmith. Contributed to production-facing loan, savings, and AI-powered systems with a focus on clean APIs, automation, reliability, and practical delivery.

CORE COMPETENCIES

Languages	Python, Java, TypeScript, JavaScript
Frameworks	Spring Boot, Spring WebFlux, FastAPI, NestJS
AI & Agents	LangChain, LangGraph, LangSmith, OpenAI API, RAG, tool calling, prompt engineering
Data & Messaging	PostgreSQL, MySQL, Chroma, Apache Kafka, Prisma ORM
Platform	Docker, Kubernetes, Jenkins, ArgoCD, Git, Postman
Ways of Working	Agile/Scrum, mentoring, technical writing, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Associate Software Engineer | Axiata Digital Labs, Sri Lanka

May 2025 - Present

Backend engineering for fintech lending and savings platforms, with additional contribution to enterprise GenAI initiatives.

- Build and maintain backend services that support lending and savings workflows across loan creation, retrieval, auto-collection, settlement, and cancellation flows.
- Design and integrate APIs across business-critical loan lifecycle stages while improving maintainability through refactoring and service decomposition.
- Optimize SQL queries and service behavior to improve response times, stability, and operational visibility in production-facing systems.
- Implement event-driven workflows using Kafka and webhook integrations to support automation, asynchronous processing, and external system coordination.
- Contribute to CI/CD delivery and release readiness in collaboration with QA, product, and platform teams.
- Mentor junior engineers during debugging, implementation reviews, and day-to-day delivery work.

GenAI / AI contribution highlights:

- Co-designed a Retrieval Augmented Generation subsystem for enterprise core-banking use cases.
- Developed document ingestion and retrieval pipelines covering normalization, chunking, embeddings, and vector search.
- Used LangChain to orchestrate LLM execution and exposed backend functionality through REST APIs aligned with security and compliance expectations.

Software Engineering Intern | Axiata Digital Labs, Sri Lanka

Jan 2024 - Sep 2024

Supported delivery of backend modules for fintech lending systems and assisted with deployment-oriented engineering work.

- Worked on backend development and integration tasks for loan processing modules, including core-banking-related services and eKYC flows.
- Improved API response times by profiling bottlenecks, refining queries, and assisting with code-level refactoring.
- Helped containerize services and supported deployments using Docker and Kubernetes.

SELECTED PROJECTS

The following projects reflect a mix of enterprise delivery, applied GenAI work, and research-oriented engineering.

Data Synthesizer | *Axiata Digital Labs*

- Designed enterprise-oriented RAG ingestion and retrieval pipelines with semantic-aware chunking, embeddings, vector search, and LangChain orchestration.
- Worked on practical backend patterns for preparing structured and unstructured content for downstream AI retrieval use cases.

Tech: Python, LangChain, Chroma

Multi-agent HR AI Assistant | *Personal Project*

- Built a multi-agent HR assistant using OpenAI, LangChain, FastAPI, and PostgreSQL.
- Implemented safe read-only SQL access, supervisor-based routing, clarification handling, and database-backed answer generation through tool calling.

Tech: OpenAI, LangChain, FastAPI, PostgreSQL

Blog Generator ReAct-type Agent | *Personal Project*

- Implemented a LangGraph-based ReAct workflow using Groq for blog generation with explicit state transitions and deterministic tool execution.
- Added bounded recursion, intermediate reasoning management, and full execution tracing through LangSmith.

Tech: LangGraph, Groq, LangSmith, Python

Decap Hire Purchase | *Axiata Digital Labs*

- Implemented APIs and workflow support for automated collections, early settlement, and cancellation processes.
- Integrated event-driven behavior to improve automation across fintech service flows.

Tech: Spring Reactive, Kafka, PostgreSQL, Mambu, Jumio, Splunk, Jenkins, ArgoCD

Boost SME | *Axiata Digital Labs*

- Delivered onboarding and revolving-credit-related APIs and supported eKYC integration work.
- Resolved service issues affecting statement-related behavior and backend stability.

Tech: Spring Reactive, Kafka, PostgreSQL, Mambu

AxoCapital SaaS | *Axiata Digital Labs*

- Worked on multitenancy and tenant-isolation features for a SaaS platform.
- Contributed to centralized authentication and audit-oriented backend capabilities.

Tech: Spring Boot, Keycloak, PostgreSQL

AI-Powered Reddit Moderation | *University of Moratuwa*

- Created a BERT-based classifier for off-topic and inappropriate content detection.
- Combined language modeling with novelty-oriented features for moderation-related predictions.

Tech: Python, BERT, scikit-learn

CERTIFICATIONS

Quickstart: LangSmith Essentials | [LangChain](#)

Issued Apr 2026 | Credential ID: vd55h4nxdp

- Focused on core LangSmith workflows for tracing, evaluation, and continuous improvement using live production data.
- Strengthened practical understanding of observability and feedback loops for agent engineering.

Quickstart: LangSmith Agent Builder | [LangChain](#)

Issued Mar 2026 | Credential ID: as5soeqwvp

- Covered building, debugging, and evaluating AI agents using LangSmith.
- Included tracing, tool-call inspection, reasoning-path debugging, experiment tracking, and output evaluation.

Project: Ambient Agents with LangGraph | [LangChain](#)

Issued Apr 2026 | Credential ID: lbrbpxlqdp

- Hands-on project on ambient-agent patterns using LangGraph.
- Explored event-driven workflows, human approval flows, state management, tool calling, and long-running agent patterns.

Foundation: Introduction to LangChain - Python | [LangChain](#)

Issued Mar 2026 | Credential ID: jtlromy7bq

- Covered foundational LangChain concepts for building LLM-powered Python applications.
- Included prompt templates, chains, tool integration, memory, and working with multiple LLM providers.

Deep Agents Certification - AI Agent Engineering | [LangChain](#)

Issued Mar 2026 | Credential ID: mao7fnqc8e

- Focused on planning-capable AI systems, long-context reasoning, and multi-agent coordination.

Additional Certificates | [Great Learning Academy](#) / [Simplilearn](#)

Issued 2024

- OOP in Java, Data Structures and Algorithms in Java (Intermediate), Introduction to Database and SQL, and Introduction to Kubernetes.

EDUCATION

University of Moratuwa

B.Sc. (Hons) Information Technology

2019 - 2025 | Second Class Upper Division

WRITING AND PROFESSIONAL PROFILES

- Regular contributor to Axiata Digital Labs engineering writing and personal Medium posts.
- Writing topics include fintech microservices, API design patterns, RAG integration, Kubernetes deployment, and applied machine learning use cases.
- Notable article: 'Building My First Production-Grade RAG System: An Engineer's Entry into Enterprise LLM Integration' (Dec 2025).

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LANGUAGES

- Sinhala and Tamil (Native)
- English (C2 Professional Proficiency)

REFEREES

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